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MTF236 - Road Vehicle Aerodynamics - VT2025
Wind Tunnel Project on Bus Aerodynamics

Group 2

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1 Introduction

Aerodynamics plays a crucial role in modern vehicle design, influencing *fuel efficiency, stability, and overall performance*. In the case of large commercial vehicles such as *buses*, aerodynamic optimization is particularly important due to their *large frontal area, bluff body shape, and sensitivity to crosswinds*. Reducing aerodynamic drag can significantly improve fuel economy, while minimizing *undesirable lift forces and side forces* enhances vehicle stability and passenger comfort.

Wind tunnel testing provides a *controlled and repeatable environment* to study vehicle aerodynamics and assess the impact of design modifications. By measuring forces such as *drag (F_D), lift (F_L), and side forces (F_S)*, as well as analyzing *flow behavior through smoke and tuft visualization*, engineers can identify opportunities to optimize vehicle shape and improve performance.

This project focuses on **analyzing and optimizing the aerodynamics of a bus model** using **wind tunnel experiments**. The study consists of the following key steps:

1. **Pre-study phase** – Computational Fluid Dynamics (CFD) analysis of the baseline bus aerodynamics to identify problem areas.
2. **Wind tunnel testing** – Experimental evaluation of the bus model under different flow conditions, including a *Reynolds sweep* and *yaw angle testing*.
3. **Aerodynamic modifications** – Implementation of *bus-specific add-ons* such as *boat-tailing, vortex generators, wheel arch-covers and rounded fronts* to reduce drag and mitigate crosswind effects.
4. **Post-processing and analysis** – Calculation of *aerodynamic coefficients* (C_D , C_L , C_{LF} , C_{LR}) and comparison between *baseline and modified configurations*.

The primary objective of this study was to evaluate how **aerodynamic modifications influence drag and stability** in a *controlled wind tunnel environment*. By quantifying the impact of these modifications, the findings provide valuable insights into strategies that can enhance the efficiency and stability of real-world buses.

2 Theoretical Background

Aerodynamics heavily influence vehicle dynamics, affecting *drag*, *lift*, *stability*, and *energy efficiency*. The forces acting on a vehicle in motion can be categorized into *aerodynamic drag (resistance)*, *lift forces*, and *moments* that influence vehicle performance and handling. This section presents the fundamental aerodynamic principles relevant to the wind tunnel testing conducted in this project.

2.1 Wind Tunnel Testing and Reynolds Number Effects

Wind tunnels are essential tools for replicating *real-world airflow conditions* in a controlled laboratory environment. They enable precise measurement of aerodynamic forces and moments acting on scaled models. In this project, all tests were conducted in the Chalmers L2 wind tunnel—a *closed-circuit, low-speed facility*—with a test section measuring $1.25\text{ m} \times 1.8\text{ m} \times 3.0\text{ m}$ and a maximum free-stream velocity of 63 m/s . The tunnel allows for accurate assessment of forces such as *drag* (C_D) and *lift* (C_L , C_{LF} , C_{LR}) using a multi-axis force balance.

Wind Tunnel Theory

The wind tunnel operates by generating a controlled and uniform airstream over a stationary model, simulating the relative motion experienced during real driving. As illustrated in Figure 1, the L2 tunnel employs a *closed-circuit* layout in which air is continuously recirculated, ensuring stable flow conditions and energy efficiency.

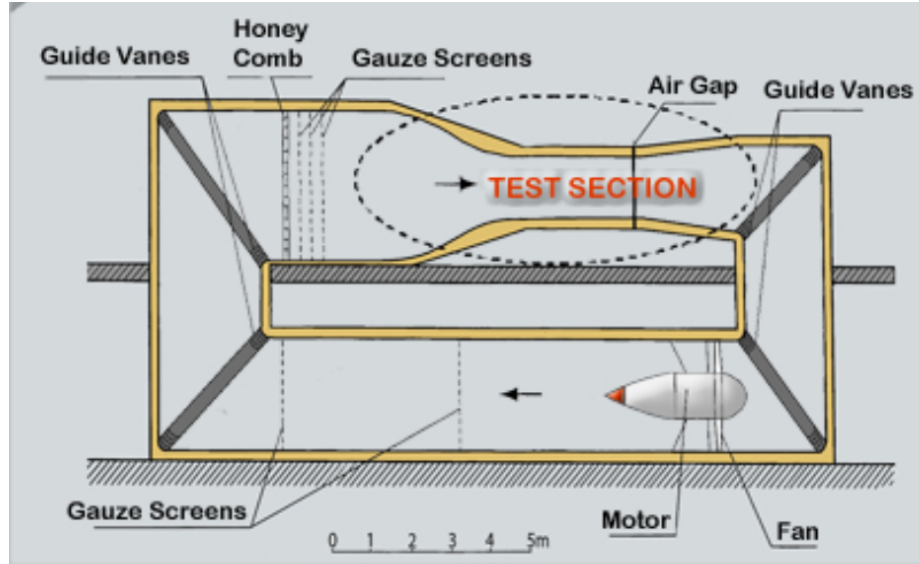


Figure 1: Schematic of Closed-Circuit Wind Tunnel Used for Testing [1]

Key components of the tunnel system include:

- **Fan and Motor:** Located downstream, these generate the primary airflow through the tunnel loop.
- **Guide Vanes:** Positioned at each corner, they help the air navigate bends with minimal pressure loss and turbulence.
- **Honeycomb and Gauze Screens:** These flow-conditioning elements straighten the airflow and eliminate residual turbulence before it enters the test section.

- **Test Section:** This is the core measurement zone where the model is mounted. An *air gap* between the floor and model platform is included to better replicate open-road boundary conditions and reduce wall effects.

This configuration ensures a well-conditioned flow that closely resembles full-scale driving conditions, making it possible to investigate aerodynamic phenomena such as flow separation, wake formation, and vortex behavior. Such effects are particularly relevant when testing bluff-body vehicles like buses.

Below the wind tunnel test section is a balance that enables the measurement of forces, moments, and controls the yaw angle due to its ability to rotate. The balance can be seen in Figure 2 and is placed in a way that allows the bus model to be mounted directly on top of it.

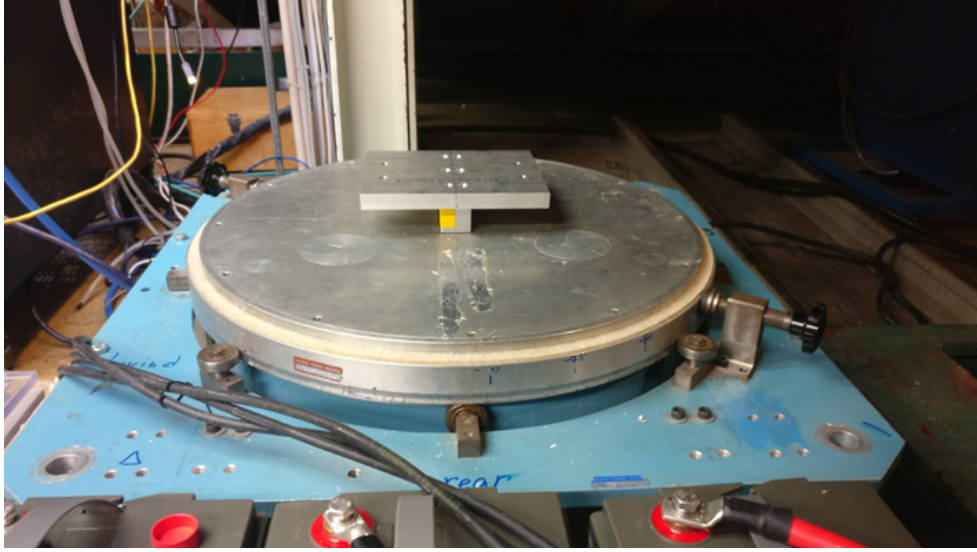


Figure 2: Measuring equipment inside the wind tunnel [1]

Reynolds

Since the wind tunnel operates at lower Reynolds numbers than those experienced by full-scale vehicles, it is necessary to account for scale effects. The *Reynolds number* (Re) is a dimensionless quantity used to characterize flow regimes and is defined as:

$$Re = \frac{\rho UL}{\mu} \quad (1)$$

where:

- ρ is the air density [kg/m^3],
- U is the flow velocity [m/s],
- L is the characteristic length [m], such as vehicle length or wheelbase,
- μ is the dynamic viscosity of air [$\text{Pa}\cdot\text{s}$].

To assess the influence of Reynolds number on aerodynamic forces, a *Reynolds sweep* is performed by varying the tunnel speed and recording the corresponding changes in drag and lift coefficients. This allows the extrapolation of wind tunnel results to full-scale operating conditions.

Free body diagram

Free-body diagrams were created to visualize how the forces F_L , F_Y , and F_D act on the body of the bus. From Figure 3, the force of the side wind generates a lateral force acting on the bus. From Figure 4, the force of the headwind generates both drag and lift forces acting on the bus. Additionally, these forces generate corresponding moments, such as yaw moment (M_{Yaw}), roll moment (M_{Roll}), and pitch moment (M_{Pitch}).

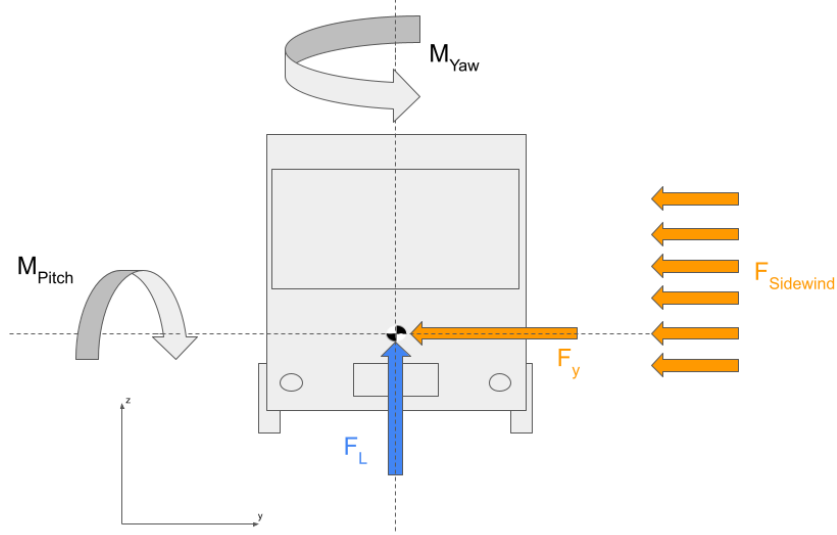


Figure 3: Free body diagram of the bus front

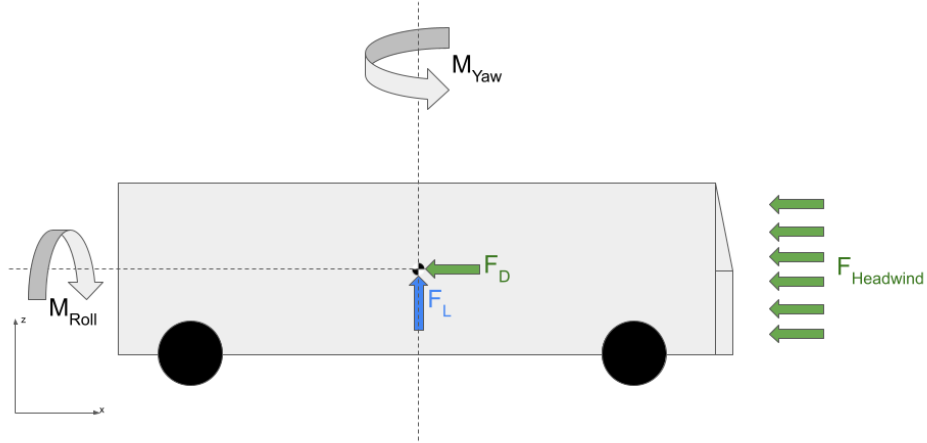


Figure 4: Free body diagram of the bus side

2.2 Aerodynamic Forces and Moments

The forces acting on a vehicle include:

- **Drag Force (F_D):** The resistance opposing vehicle motion, primarily due to *pressure drag* and *skin friction*.
- **Lift Force (F_L):** The vertical force generated by airflow over the vehicle, influencing stability. Lift is separated into:
 - *Front Axle Lift (C_{LF})* – affects steering response and high-speed stability.
 - *Rear Axle Lift (C_{LR})* – influences traction and handling. Which is regarded as less important in bus travel.
- **Side Forces and Yaw Moments:** These determine how *crosswinds* affect stability.

The *drag coefficient* (C_D) and *lift coefficient* (C_L) are dimensionless parameters used to compare aerodynamic efficiency across different vehicle shapes and speeds:

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2 A} \quad (2)$$

$$C_L = \frac{F_L}{\frac{1}{2}\rho U^2 A} \quad (3)$$

where A is the frontal area of the vehicle.

2.3 Aerodynamic Modifications and Optimization

Since our group is working with a **bus**, the primary objective is to *reduce pressure drag* and improve aerodynamic efficiency while maintaining passenger and cargo space. Unlike passenger cars or racing cars, where the focus is on *downforce and cornering stability*, buses have *large frontal areas* and *box-like shapes* that cause significant aerodynamic drag.

Common **bus-specific modifications** tested in the wind tunnel include:

- **Front-end optimization:** Increasing *corner radius* and adjusting *windscreen angles* to reduce stagnation pressure and minimize flow separation.
- **Rear-end optimization:** Implementing *boat-tailing* or *roof extensions* to reduce wake turbulence and increase base pressure.
- **Underbody modifications:** Addressing airflow around *exposed wheels* and *side skirts* to reduce turbulence.
- **Yaw angle testing:** Evaluating the impact of *crosswinds* on bus stability and proposing *aerodynamic enhancements* to mitigate side forces.

By analyzing these modifications in the wind tunnel, we can quantify their impact on *drag reduction* and determine the most effective strategies for improving the bus's aerodynamic performance. The individual modifications are discussed in greater detail in the next section.

3 Methods

3.1 Pre-study

There was an opportunity to study a completed CFD simulation of a modeled bus. In this session, it was noted that from Figure 5:

- The front of the bus was too flat and square, causing high stagnation of air at the front, and thus high pressure, increasing drag.
- The rear of the bus was also similarly boxy, resulting in the formation of a large wake.
- The flow on the side surfaces of the bus was mostly attached, but appeared to detach at some regions around the rear wheels. These three observations can be seen in Figure 5.

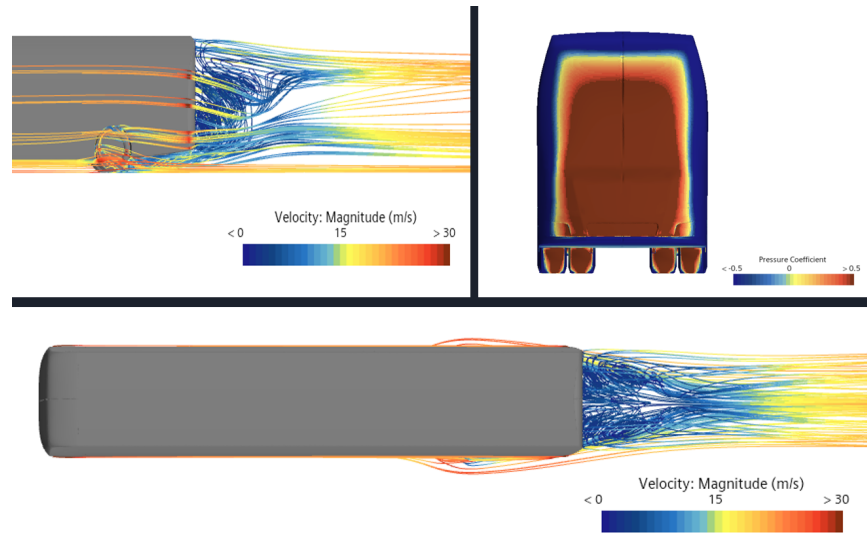


Figure 5: Results From The Pre-Study

Accordingly, it was decided that, in order to reduce drag, the following solutions should be attempted:

- Reduce the area of stagnation at the front of the bus by making a more rounded front.
- Delay flow separation at the rear as long as possible, and create the smallest possible wake.
- Reduce flow separation around the wheels if possible.

3.2 The model and its parts

A 1:18 bus model was provided, along with some pre-made attachments for the nose and tail. In addition to a regular "base" nose and tail as seen in Figure 6, a bullet train-like nose and a boat tail were among the given parts, seen mounted on the front and rear in Figure 7.

For some tests, the baseline nose piece was modified with modelling clay to resemble the snout of a beluga whale (*Delphinapterus leucas*), with two small hemispherical protrusions, as seen in Figure 8. This was to emulate the work of Arabaci & Pakdemirli, 2023 [2].



Figure 6: The baseline model



Figure 7: The model with a bullet-train nose, boat tail, and other modifications

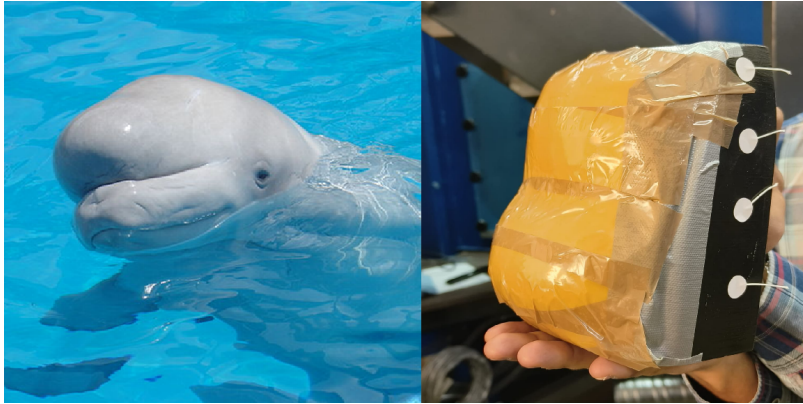


Figure 8: A beluga whale (left) and the inspired nose modification (right)

In addition to these, certain smaller parts commonly used as aerodynamic modifications were also prepared; they are visible in Figure 7, while their real-life inspirations are visible in Figure 9.

- Diffusers are used to delay the underbody flow separation by releasing the fast airflow under the vehicle into the slower ambient air more gently. Diffuser modifications are usually comprised of vertical slats, and such a model was implemented with cardboard and duct tape on the bottom of the boat tail.
- Vortex generators are small fins that are placed in regular intervals to generate small, controlled vortices along the surface of a flow, which reduce the size of the boundary layer, thus preventing early flow separation. In our model, this was implemented with duct tape.
- To prevent the disturbance of the flow around the wheels, they can be covered by flat body panels, as seen, for instance, in the Brisbane Metro articulated buses. We implemented this

by taping over the front wheels with cellophane.



Figure 9: Clockwise from top: Diffuser slats used in a racing car; vortex generators on the leading edge of an airplane wing; wheel covers on a bus in the Brisbane Metro system

3.3 Wind tunnel testing

The parts discussed above were attached to the model in various configurations, as described in Table 1.

Table 1: Chosen Design Modifications

Configurations	Modifications
Config 1	Bullet nose [BN], boat-tail rear with bottom diffusers [BT+D], wheel covers (front) and arches (Rear) [WC], vortex generators [VG]
Config 2	Bullet nose, baseline rear [BR], wheel covers (front) and arches (rear), vortex generators
Config 3	Beluga front [Bel], base rear, wheel covers (front) and arches (rear), vortex generators
Config 4	Beluga front, boat-tail rear with bottom diffusers, wheel covers (front) and arches (rear), vortex generators
Config 5	Beluga front, boat-tail rear [BT] only
Config 6	Baseline front and baseline rear (<i>reference</i>)

The wind tunnel at Maskinhuset, Campus Johanneberg, is a closed-circuit tunnel with open-loop control of its velocity and the possibility to vary the yaw angle of the model in the windstream. Since the modifications attempted on the central piece of the bus model were largely superficial, this part of the model was bolted down in advance to the turntable, which is connected to the balances. This

way, the nose and tail pieces could be removed and modified between tests, while the modifications attached to the central part were just removed when no longer needed.

First, a Reynolds sweep was performed on the bus model in its baseline configuration, by increasing the wind speed from zero to $50m/s$. The drag coefficient variation was noted down and $35m/s$ was the speed at which the increase in drag force became linear with speed, indicating that the drag coefficient value had settled (see Figure 10). This was therefore the wind speed chosen for the remainder of the tests.

Each configuration was placed at yaw angles of 0° , 2.5° , 5° and 10° in the tunnel, and the lift and drag readings recorded.

3.4 CFD simulation

It was deduced from the drag force alone that Configuration 5, with the beluga nose and the boat tail, was the most superior. Some observations were taken with the smoke machine and observing the tufts, but it was desired to have a deeper insight into why exactly it was so much better. Thus, a very basic CFD simulation of around 600 iterations and a fairly rough mesh was performed on both the baseline geometry and the best-performing geometry, though it was not the main goal of this project. The intention was solely to visualise the flow and the stagnation pressure areas, and not for accurate numerical results.

4 Results and Discussions

In the wind tunnel, a balance under the test bed measures the forces acting on the bus which are used to calculate and plot different force and drag coefficients. These will be shown and explained in detail in the following section.

4.1 Reynolds Sweep

As previously mentioned, the first test is a Reynolds sweep which means the air flow velocity is swept within a specified range. In the drag force sweep plot shown in Figure 10a, the drag force has a squared relation with velocity ($F_D \propto u^2$) as derived from Equation 2. Regarding the drag coefficient shown in Figure 10b, it initially exhibits a high value at low velocities but gradually decreases as velocity increases, eventually stabilizing to a constant value. This trend can be attributed to two key factors:

Reynolds Number Variation: At low velocities, the Reynolds number (Re) is relatively low, resulting in dominant viscous forces and increased skin friction drag. As velocity rises, the Reynolds number increases, leading to a shift where inertial forces become more significant. This transition reduces the drag coefficient until it reaches a stable value.

Flow Regime Transition: At higher velocities, the flow becomes turbulent. Turbulent flow delays flow separation, thereby reducing pressure drag and contributing to a lower and more stable drag coefficient.

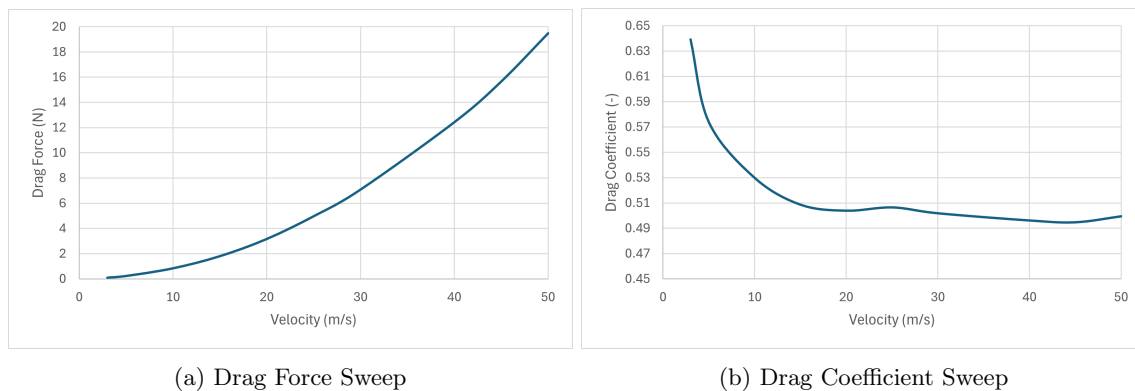


Figure 10: Reynolds Sweep

For this analysis, only drag force variations are considered, while lift forces (positive and negative lift) are neglected due to the large weight of the bus and non-high performance needs (down force). Considering a long tour highway bus which operates continuously above 60 km/hr where aerodynamic forces start to act, generation of positive lift forces will have a negligible impact on the overall stability due to bus's heavy weight. For a city buses, they primarily operate at low speeds. Hence, lift effects are insignificant in this context.

4.2 Yaw Sweeps

Yaw sweeps are used to assess the driving stability of vehicles with the presence of cross-winds. It is especially important to check this on longer vehicles such as buses and trucks since the exposed area is significantly larger and can pose a safety risk. To demonstrate this, a visual representation of a

cross-wind on a bus is shown in Figure 11, from which a formula for the total area exposed normal to the flow of air is derived.

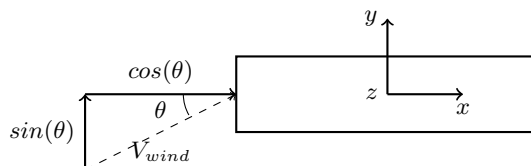


Figure 11: Cross-Wind Effect on Bus

The derivation of the exposed area is as follows,

$$A_{f-yaw} = A_f \cos(\theta) + A_s \sin(\theta), \quad (4)$$

where A_f is the frontal area of $0.0262m^2$ (common to all configurations), A_s is the side area of $0.146m^2$ (found from CAD model), and θ is the yaw angle. The newly calculated frontal areas are summarized in Table 2, where it is seen that the exposed area increases with yaw angle as expected, since more of the large side area gets exposed.

Table 2: Frontal Areas for Yaw Angle Sweep

Yaw Angle (deg)	Frontal Area (m^2)
0	0.0262
2.5	0.0298
5	0.0333
10	0.0404

It should be noted that this increased exposed area is only used to demonstrate the importance of yaw sweeps for long vehicles and **not** used to determine the drag coefficient C_d from Equation 2. The A substituted in that equation remains fixed at $0.0262m^2$ for the purposes of our calculations.

In Figure 12, we see how the drag coefficient varies with yaw angle for the six different configurations. The base model, Figure 6, exhibits the highest drag coefficient, which increases with yaw angle. This can be primarily attributed to its fully flat frontal surface with minimal edge radii, leading to a larger stagnation area and higher stagnation pressure. Furthermore, it can be noted that all modifications have improved the drag coefficient. The approximate changes in drag coefficient C_d in comparison to the baseline model due to each addition are presented in Figure 13.

Observing the different trend lines in Figure 12, it can be seen that the boat tail (green and orange lines) outperforms the base rear (purple) due to its smaller area and wake. Additionally, the bullet nose (dotted line) outperforms the beluga (solid lines) front when comparing to the same rear attachment used (comparing within purple and within green lines).

Due to the impracticality of a nose like a bullet train for a road vehicle, the beluga nose and the boat tail are strong contenders for meaningful real-life additions, along with wheel covers. The model with those three additions exhibited a C_d of 0.295, an improvement of around 210 counts over the baseline model which had a C_d of around 0.505.

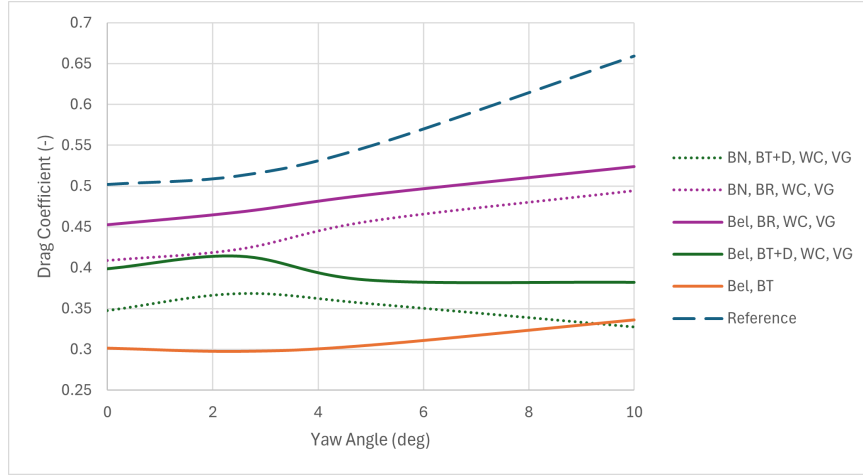


Figure 12: Drag Coefficient Yaw Angle Sweep

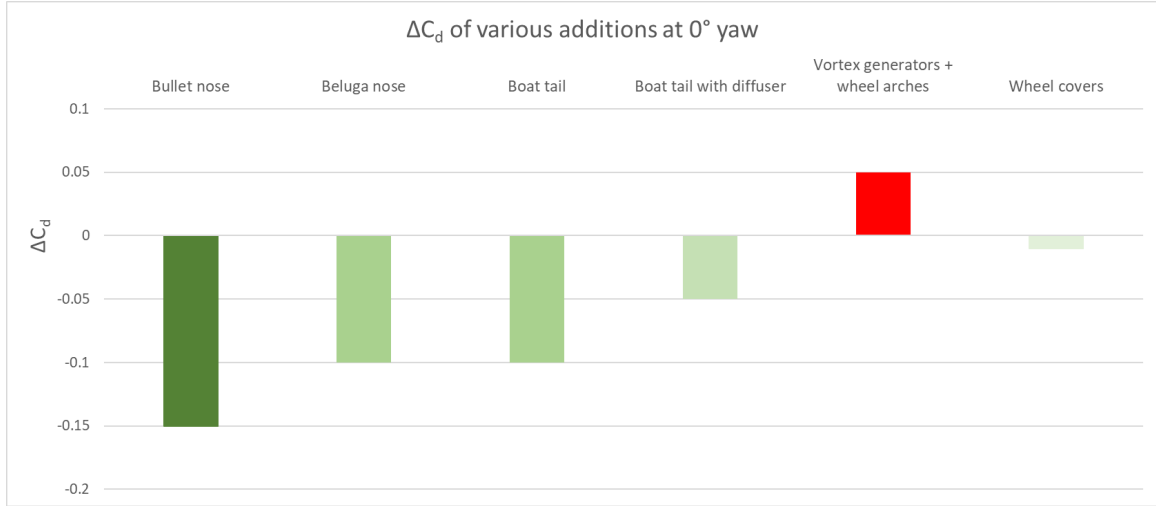


Figure 13: Change in drag coefficient w.r.t baseline model due to different additions

The attachments made of cardboard and duct tape appeared, unfortunately, to add drag. This was likely because of their imprecise dimensions and poor build quality.

A comparison of the two best-performing configurations highlights key aerodynamic trade-offs. Ideally, a combination of aerodynamic devices, smooth surface finish, reduced frontal area, and boat tailing should yield the lowest drag coefficient. However, practical limitations such as cost and aesthetics restrict this full implementation. Among the six tested configurations, Configuration 1 (Figure 7) featured a bullet-nose front, boat tail, bottom diffuser, front axle wheel covers, rear wheel arches, and vortex generators, whereas Configuration 5 (Figure 14) incorporated only a Beluga-shaped front and a boat tail. As shown in Figure 3, the drag coefficient (C_d) for Configuration 1 initially increases up to a yaw angle of 2.5° , then gradually decreases. In contrast, Configuration 5 starts with a lower C_d , remains nearly constant until 5° yaw, and then increases gradually. While Configuration 5 exhibits a slightly higher C_d at maximum yaw, it maintains a lower value throughout most of the range.

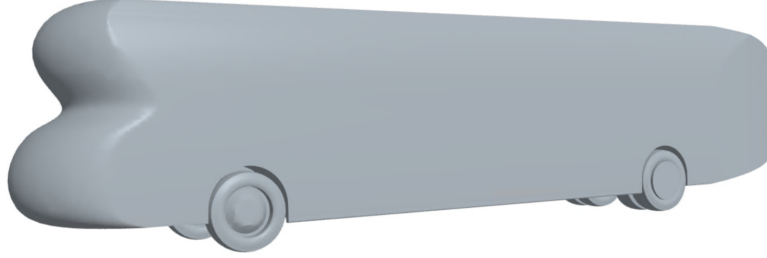


Figure 14: The best-performing model, Configuration 5 - beluga nose, boat tail

The results from the smoke test in the wind tunnel and observing the movement of the tufts suggest that the beluga nose-boat tail combination's success was largely because of a smaller wake in the rear, plus a reduced frontal stagnation area. The visualizations from the CFD study confirm this, as shown Figure 15 and Figure 16.

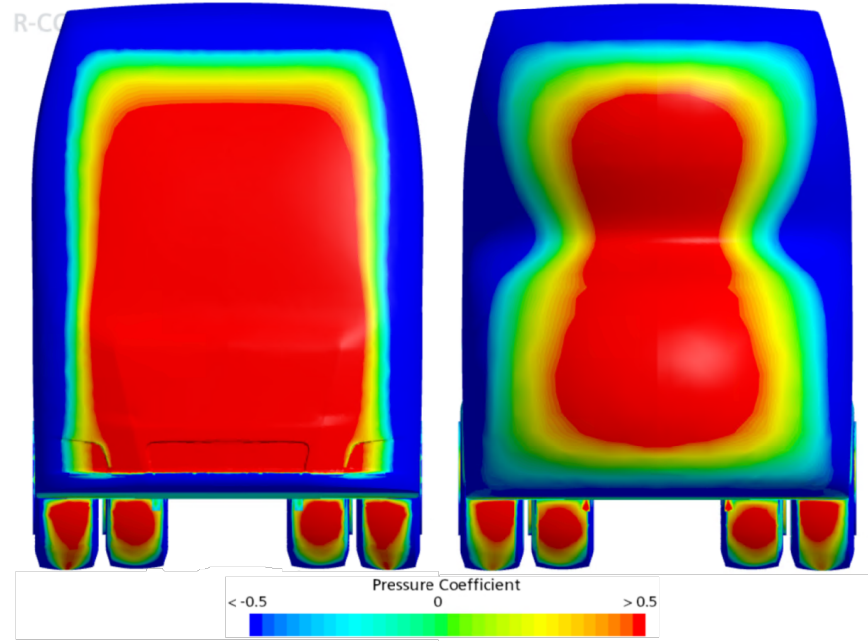


Figure 15: Comparison of stagnation pressure regions on the baseline model (left) and the beluga-nosed model (right)

It must be noted that these simulations were not run with the intention of precisely modeling the flow to obtain exact drag values, but just to visualize what was not possible to observe directly in the wind tunnel.

From a practical perspective, Configuration 5 is more suitable than Configuration 1, as excessive aerodynamic modifications increase vehicle weight, reducing fuel efficiency, and adding to production and maintenance costs. Additionally, the bullet-shaped nose of Configuration 1 is not aesthetically appealing or realistic to add to a road vehicle.

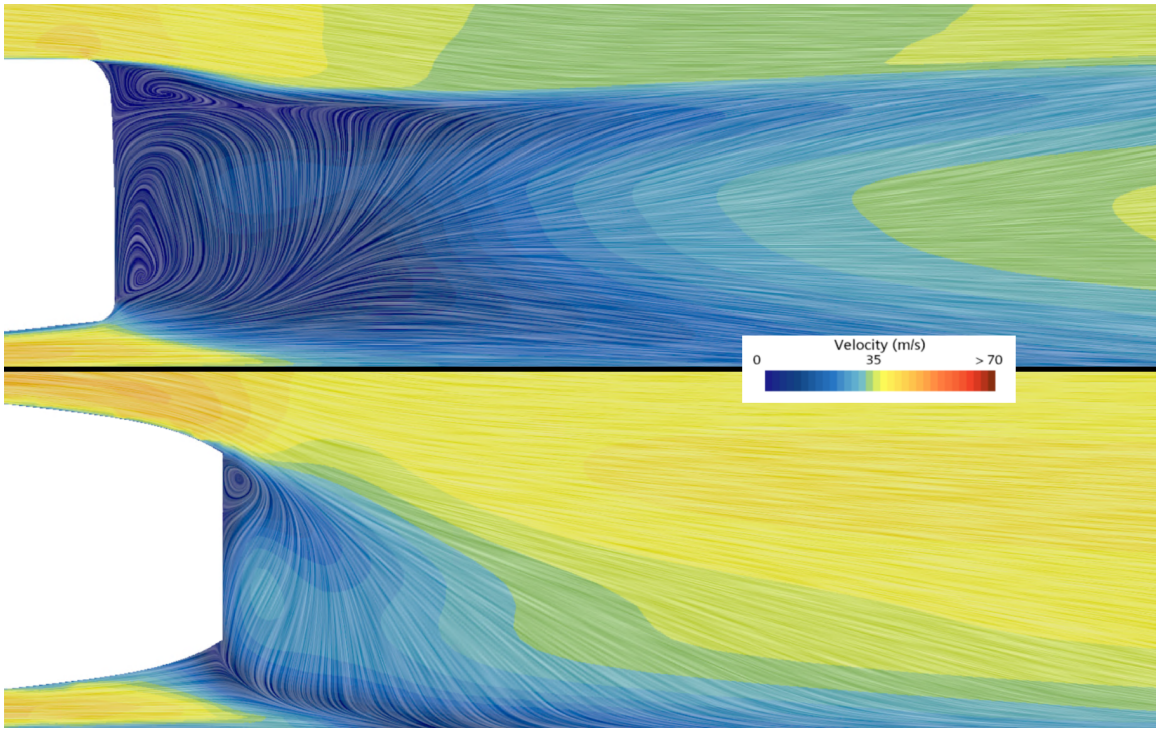


Figure 16: Comparison of the rear wake on the baseline model (top) and the boat-tail model (bottom)

5 Conclusions

The results confirm that the established theory and equations are valid for our tests.

For a bus, we are primarily concerned about improving the drag performance as it directly affects the fuel consumption. The lift force is not of much importance in such heavy vehicles.

Well-made additions such as the bullet train nose, the beluga nose, and the boat tail are very useful in reducing the drag coefficient.

The other aerodynamic devices were made of cardboard and attached via duct tape and seal tape. Although we tried our best to tape it perfectly on the surface, there were a few regions of improper taping which created rough surfaces on the bus. This might have contributed to additional drag that could not be accounted for in the results.

All attempted modifications improved the drag coefficient of the bus, but the model with just the beluga whale-like nose and the boat tail is the most realistically implementable and performed the best. This was because the curved nose reduced the stagnation area and the convergent rear resulted in a smaller wake.

The beluga nose shape itself is fairly complex, though smaller than a fully spherical front, and challenges in real life implementation include regulations, ergonomics, and manufacturing costs. A boat tail also adds length to the bus.

Further analysis with a more accurate setup - possibly with a moving ground and spinning wheels - can be used to fine-tune additions that were roughly made in this test, such as the diffuser and the vortex generators. The parts can also be additively manufactured for greater precision. In particular, the vortex generators need to be much smaller, more numerous, and more regularly spaced, and the diffuser slats need to be more precisely modelled.

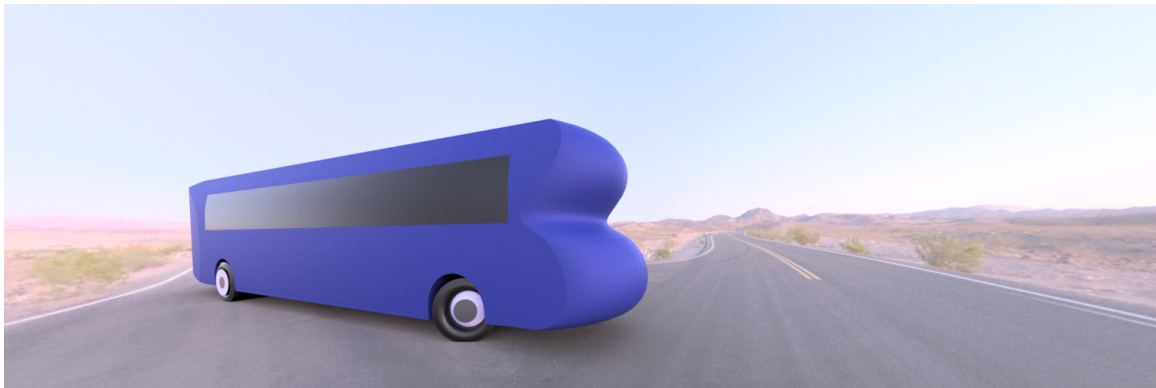


Figure 17: A rendering of the best-performing model in a real-life setting

6 Sources

References

- [1] V. Alexey, “Wind tunnel project presentation,” January 2025. Lecture slides, Road vehicle aerodynamics Course [MTF236], Chalmers University of Technology.
- [2] S. K. Arabaci and M. Pakdemirli, “Aerodynamic improvements of buses inspired by beluga whales,” *Journal of Applied Fluid Mechanics*, 2023.

Images

The images of the beluga whale in Figure 8 and the aerodynamic parts in Figure 9 are shared under Creative Commons Attribution ShareAlike licences and can be found on Wikimedia Commons. Copyright belongs to the respective photographers.

- Beluga whale: [https://commons.wikimedia.org/wiki/File:Delphinapterus leucas 3.jpg](https://commons.wikimedia.org/wiki/File:Delphinapterus_leucas_3.jpg)
- Vortex generators: [https://commons.wikimedia.org/wiki/File:TA-4SU Skyhawk cockpits.jpg](https://commons.wikimedia.org/wiki/File:TA-4SU_Skyhawk_cockpits.jpg)
- Diffuser: [https://commons.wikimedia.org/wiki/File:Corvette C6.R Atlanta esses.jpg](https://commons.wikimedia.org/wiki/File:Corvette_C6.R_Atlanta_esses.jpg)
- Brisbane Metro bus: [https://commons.wikimedia.org/wiki/File:Brisbane Metro bus testing on the Northern Busway, Brisbane, February 2024 P1001852.jpg](https://commons.wikimedia.org/wiki/File:Brisbane_Metro_bus_testing_on_the_Northern_Busway,_Brisbane,_February_2024_P1001852.jpg)

A Individual Contributions

A.1 Elin

I have been present at all group meetings and consultation sessions in addition to the work I contributed in the wind tunnel and report. During the building workshop, I worked on creating the new bus fronts with Anirudh. This mainly comprised of the beluga front as an additional 3D printed bullet nose was provided. During the wind tunnel testing, I was responsible for going inside to change the configurations, clean up after the smoke test and attach visual markers (string tassels). After the tests, I was responsible (with Bhushan) for the results which comprised of making plots from the raw data and the text write up for that section. I also helped with the PowerPoint presentation and was a presenter for our group.

A.2 Elliot

During the wind tunnel workshops I contributed to experiments regarding wheel arch covers and during the wind tunnel lab itself I took notes for the group writing down the order of tests and logged the data as a backup. For the report and presentation I mainly worked on Introduction, background and conclusion to the project in the report and created a layout and structure for the presentation with Linus.

A.3 Linus

During the wind tunnel workshops, I worked on the vortex prototype experiments and helped build the prototypes used on the bus model. In the wind tunnel lab, I took photos and videos of the tests for documentation and made sure we followed the testing schedule so we could complete as much as we could on time. For the report, I mainly contributed to the Introduction and Theoretical Background sections but also in the conclusion. Elliot and I also worked together on the layout and structure of the PowerPoint presentation to make it clear and well-organized.

A.4 Bhushan

During the wind tunnel exercise, I worked on integrating vortex generators into our model. I made a triangular shape vortex generator for the workshop and also captured images of different configurations for our reference. In the Results section, I worked on reports, making graphs of coefficient of drag vs different parameters like Reynolds number, velocity and different yaw angles. I also contributed in 3D modeling of the bus for 2 configurations namely beluga front with boat tail rear and a base model of the bus.

A.5 Ramkumar

During the wind tunnel workshop, I contributed to making the diffuser model and the extended open flaps model for the rear end (along with Himanshu). Additionally, I assisted in changing the yaw angle for different configurations and captured images and videos of the wind tunnel testing. I also attached tufts to each model and held the smoke pipe for the smoke test. Apart from the workshop, I contributed to the report (along with Anirudh) by adding the Methods section, which included CFD and experimental data as well as 3D CAD modeling.

A.6 Saket

In the post-processing workshop, I collaborated with Himanshu to analyze the simulation results of the base model and identify areas for aerodynamic improvement. During the building workshop, Elliot and I focused on optimizing the wheels by building front axle wheel covers to minimize flow separation and rear axle arches to effectively redirect airflow. In the wind tunnel lab, I worked with Elin and Anirudh to prepare the necessary attachments for testing. Apart from practical performance, I contributed to the discussion section of the report with Himanshu, interpreting the simulation graphs and determining the most aerodynamically efficient configuration. At last, I presented our project along with Anirudh, Elin and Himanshu.

A.7 Himanshu

I actively attended all group meetings and consultation sessions. For post-processing workshop, I collaborated with Saket for CFD analysis. In wind tunnel session, I collaborated with Ramkumar to make diffuser and extended open flaps. I also assisted for making yaw angle adjustments and attaching tufts on the bus surface. For report, I collaborated with Saket for writing Discussion section. I also added few points in Conclusion section personally. Finally, I collaborated with Saket, Anirudh, and Elin to present the presentation.

A.8 Anirudh

In the post-processing workshop, I contributed ideas such as the wheel covers and the beluga nose. I collaborated with Elin to make the beluga nose, for which I supplied the modelling clay. On testing day I operated the wind tunnel itself, controlling the wind speed and logging the force data from the balance. In this report I mainly collaborated with Ramkumar to write the section on methods and to perform the CFD simulation. Finally, I presented our work on presentation day alongside Elin, Himanshu, and Saket.